



IIT BOMBAY

GNR 638 : MACHINE LEARNING FOR REMOTE SENSING - II

INDIAN INSTITUTE OF TECHNOLOGY, BOMBAY

ELECTRICAL ENGINEERING

Theory Assignment 1

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1 Problem Statement

Question: Prove that the difference of two convex functions is not necessarily convex.

2 Theory and Explanation

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is called **convex** if for all $x_1, x_2 \in \mathbb{R}$ and for any $\lambda \in [0, 1]$, we have:

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2).$$

This property implies that the **second derivative** of $f(x)$, if it exists, must satisfy:

$$f''(x) \geq 0, \quad \forall x \in \mathbb{R}.$$

If $f(x)$ and $g(x)$ are convex functions, their second derivatives satisfy:

$$f''(x) \geq 0, \quad g''(x) \geq 0, \quad \forall x.$$

We define their difference as:

$$h(x) = f(x) - g(x).$$

Our goal is to show that $h(x)$ is not necessarily convex, meaning that $h''(x)$ may not always be non-negative.

3 Counterexample: A Detailed Step-by-Step Proof

To demonstrate this, we provide a counterexample with two convex functions whose difference is non-convex. Let:

$$f(x) = x^2 \quad (\text{convex})$$

$$g(x) = e^x \quad (\text{convex})$$

Define their difference as:

$$h(x) = f(x) - g(x) = x^2 - e^x$$

To analyze the convexity of $h(x)$, we examine its second derivative:

$$\text{First derivative: } h'(x) = 2x - e^x$$

$$\text{Second derivative: } h''(x) = 2 - e^x$$

Analyzing the sign of $h''(x)$:

- At $x = 0$: $h''(0) = 2 - 1 = 1 > 0$ (convex)
- At $x = 1$: $h''(1) = 2 - e \approx -0.718 < 0$ (concave)

Since $h''(x)$ changes sign from positive to negative as x increases, $h(x)$ is not convex over $\mathbb{R}$. Specifically:

- Convex for $x < \ln(2)$
- Concave for $x > \ln(2)$

Conclusion: The difference $h(x) = x^2 - e^x$ is not convex everywhere, proving that the difference of two convex functions may be non-convex.

The difference of two convex functions is not necessarily convex.
